CS 499

Professional Self-Assessment

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10/18/26

Throughout my time in the Computer Science program at Southern New Hampshire University, I have developed the technical knowledge, problem-solving abilities, and professional skills needed to pursue a career in software development. My coursework has given me experience with programming languages, software engineering practices, algorithms and data structures, databases, application security, and the software development process. More importantly, it has taught me how to approach unfamiliar problems, evaluate possible solutions, and improve existing systems. As I approach the completion of my degree, my ePortfolio represents both the skills I have developed and my ability to apply them to practical software projects.

One of the most valuable parts of my education has been gaining experience with different types of software development. I have worked with Java, Python, C++, JavaScript, TypeScript, C#, SQL, and several related frameworks and development tools. These technologies have allowed me to explore different areas of computer science rather than focusing on only one type of application. In CS 330, I worked with OpenGL to develop a three-dimensional graphical scene, which strengthened my understanding of rendering and graphics programming. In CS 360, I developed an Android application that incorporated device-sensor functionality. In CS 340, I worked with MongoDB and a Python Dash dashboard to manage and display database information. These experiences helped me understand how software development principles apply across different platforms and environments.

Software design and engineering have become particular areas of interest for me. Through CS 465, I developed Travlr Getaways, a full-stack web application built with Angular, Node.js, Express, REST APIs, and MongoDB. The project required me to understand how a frontend communicates with a backend, how data is stored and retrieved, and how authentication supports protected application functionality. My CS 499 capstone gave me the opportunity to return to that application and improve it rather than beginning another project from scratch. Working with an existing codebase required careful planning because new functionality needed to integrate with the application without breaking features that already worked. This experience reinforced the importance of modular design, maintainable code, documentation, testing, and evaluating the effects of changes across an entire system.

Algorithms and data structures have also influenced the way I approach programming problems. In CS 370, I worked with deep Q-learning to train an agent to solve the CartPole problem. That project introduced me to reinforcement learning and demonstrated how algorithms can improve through repeated training and evaluation. In my capstone, I applied a different set of algorithmic concepts by creating a trip-ranking feature for Travlr Getaways. I implemented weighted scoring, used a JavaScript Map to organize review information, developed a custom merge sort for complete rankings, and used a min-heap to identify the highest-rated trips. Comparing the O(n log n) complexity of merge sort with the O(n log k) processing complexity of Top-K selection helped me understand the importance of choosing an algorithm based on the actual requirements

of a problem. Automated tests also allowed me to verify the ranking calculations and ordering behavior instead of relying only on visual results.

My experience with databases has taught me that effective data management involves more than storing information. Through coursework involving SQL and MongoDB, I learned how database structures, relationships, queries, and validation affect an application's reliability and performance. During the database enhancement of my capstone, I strengthened Mongoose schema validation, introduced compound indexes, and used a unique database constraint to prevent duplicate reviews. I also implemented MongoDB aggregation pipelines to calculate review counts, category averages, and weighted scores directly within the database. Using MongoDB's query explanation tools helped me evaluate whether indexes were supporting the intended queries. This work demonstrated the value of considering database efficiency and data integrity as part of the overall software design rather than treating the database as a separate component.

Security has become another important part of my development process. In Travlr Getaways, I worked with JSON Web Token authentication, customer and administrator roles, and backend authorization checks. Adding customer reviews showed me why identifying an authenticated user is not the same as determining what that user should be allowed to do. I implemented role-based restrictions for administrative operations and ownership checks to prevent users from editing or deleting reviews belonging to other people. I also strengthened input validation and database constraints to protect application data. These improvements reinforced the importance of designing software with security in mind from the beginning and enforcing permissions on the backend instead of relying only on restrictions in the user interface.

My education and professional experience have also strengthened my communication and collaboration skills. In my role as an overnight team lead, I have been responsible for coordinating work, communicating expectations, resolving problems, and adapting to changing priorities while working with others toward shared goals. Although this work is outside the software development field, it has helped me become more comfortable taking responsibility, managing competing priorities, and communicating with people who have different responsibilities and perspectives. I believe these skills transfer directly to working on software development teams, where collaboration and understanding stakeholder needs are essential. My computer science coursework has also required me to explain technical concepts through written assignments, project documentation, and presentations. My CS 499 code review and enhancement narratives, in particular, challenged me to explain my design decisions, identify areas for improvement, and communicate the reasoning behind technical changes.

Outside of coursework, I have continued developing my programming skills through personal projects. One of my ongoing interests is game development using Unity and C#. I have been working on a survival game that allows me to explore gameplay programming, modular

architecture, and the challenges of organizing larger software systems. Personal development projects are valuable to me because they provide opportunities to experiment with ideas, solve problems independently, and learn technologies beyond those required for a particular assignment. They have also helped me recognize that software development requires continuous learning as tools, frameworks, and design practices evolve.

My CS 499 ePortfolio brings these experiences together through the enhancement of Travlr Getaways. I selected this application because it provided a strong foundation for demonstrating progress across multiple areas of computer science. My first enhancement focused on software design and engineering through a customer review system, improved authentication and authorization, and integration between frontend and backend components. My second enhancement focused on algorithms and data structures through weighted scoring, merge sort, min-heap selection, and automated testing. My third enhancement concentrated on databases through stronger validation, indexing, uniqueness constraints, and MongoDB aggregation. Together, these enhancements demonstrate how software engineering, algorithms, database design, and security can work together to improve one application.

The work presented in this portfolio also demonstrates my progress toward the five Computer Science program outcomes. My professional experience and project planning support my ability to contribute to collaborative environments and consider the needs of different stakeholders. My code review, written narratives, and project documentation demonstrate professional communication. My ranking algorithms and testing show my ability to evaluate computing solutions and consider design trade-offs. My full-stack application enhancements demonstrate my ability to use appropriate development tools and engineering practices to create functional solutions. Finally, my authentication, authorization, validation, and database-integrity improvements demonstrate a security-focused approach to software design.

Completing the Computer Science program has helped me build a foundation for my future career while also showing me how much more there is to learn. I am particularly interested in software engineering, backend development, and full-stack application development, although I remain open to opportunities in other areas of technology. I want to continue improving my technical skills, gaining experience with professional development practices, and contributing to projects that solve meaningful problems. My ePortfolio represents the knowledge and experience I have developed so far, and I intend to continue building on that foundation through future projects and professional opportunities.